MA2031-Hints to solve problems of Assignment II

1. (a) Independent.
   (b) Dependent.
   (c) Independent.
   (d) Independent.
   (e) Dependent.
   (f) Independent, Assume some linear combination of the functions gives zero vector, substitute 3 appropriate values for 't', get three equations involving α_1, α_2 and α_3, show that zero is the only solution(there can be other methods).

2. Suppose $(ad - bc) \neq 0$(i.e., determinant of the matrix corresponding to the vectors is non-zero), then multiply by the inverse of the matrix to prove linear independence. Conversely, prove $(ad - bc) = 0$ implies linear dependence.

3. Use the fact that the number of elements in a linearly independent set is less than or equal to the number of elements in a spanning set. You know a spanning set for $\mathbb{R}^2$.

4. Trivial.

5. Choose first three independent vectors in R^3, then choose the fourth vector as non-zero linear combination of all the three vectors.

6. (a) Take any linearly independent set and add the zero vector to it.
   (b) Same as (a).

7. Use the fact that e^t is never zero and the set $\{1, t, t^3\}$ is linearly independent. alternatively substitute 3 appropriate values for 't', get three equations for which zero is the only solution

8. Assume the linear combinations of the functions is zero, multiply by $sinmt, 1 \leq m \leq n$ and integrate both sides to show that the linear combinations are just zero combinations.

9. Take a linear combination =0 and group the coefficients of the like terms. Use the fact that any subset of $\{1, t, t^2, ..., t^n\}$ is a linearly independent set.

10. (a) Use the fact that a linearly independent set with $dim(V)$ elements is a basis.
    (b) Same as (a).

11. Yes. Notice that the $(n+1)\times(n+1)$ matrix formed by the vectors is an upper triangular matrix with all diagonal entries equal to 1. Again, use the fact that a linearly independent set with $dim(V)$ elements is a basis.

12. (a) First identify the dimension of V. Then use the fact that a linearly independent set with $dim(V)$ elements is a basis.

    (b) No. The set is not linearly independent.

13. $\{(1,0,0,0,1)^T, (0,1,0,1,0)^T, (0,0,1,0,1)^T\}$.

14. Any set with two polynomials of different degrees satisfying $a+2b+3c=0$, where a is the constant term of the polynomial, b is the co-efficient of t and c is the co-efficient of t^2.

15. Add vectors so that it spans the basis $\{1, t, t^2, t^3\}$. Simplest case is to add t and t^3.

16. Yes. Linear independence proved in Assignment I. Use the fact that a linearly independent set with $dim(V)$ elements is a basis.

17. Construct three 3×3 invertible matrices that meets the requirements. Notice that the columns of an invertible matrix will form a basis.

18. Define A_{ij} to be the matrix whose ij^{th} entry and ji^{th} entry is 1 and all other entries are zero and $B_{ij},\ i\neq j$ is the matrix whose ij^{th} entry is i and ji^{th} entry is $-i$ and all other entries are zero, then the collection of all A_{ij}'s and B_{ij}'s forms a basis.

19. For $i\neq j$, let A_{ij} is the matrix whose ij^{th} entry is 1 and all other entries are zero, Let B_i is the matrix whose $ii^1.th$ entry is 1 for $1\leq i\leq(n-1)$ and nn^{th} entry is - The set containing A_{ij}'s and B_i's forms a basis.

20. $\{\begin{pmatrix} i & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & i \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}\}$.

21. –

22. $\{(0,1,0,0,0)^T, (1,0,1,0,0)^T, (1,0,0,1,0)^T, (0,0,0,0,1)^T\}$.

23. $\{(0,1,1,1,0)^T, (-1,0,0,0,1)^T\}$.

24. Name the vectors as $\{v^1, v^2, ..., v^6\}$. Do RREF to get that $\{v^1, v^2, v^4\}$ is a basis. Also note, $\{v^1, v^2, v^6\}$ is also a basis.

25. (a) n^2 for complex matrices over complex field and for real matrices over real field, $2n^2$ for complex matrices over real field.

    (b) $n(n+1)/2$.for real matrices over real field

(c) $n(n-1)/2$.for real matrices over real field

(d) n^2 for complex matrices over real field.

(e) $n(n+1)/2$ for real matrices over real field,$n(n+1)/2$ for complex matrices over complex field, $n(n+1)$ for complex matrices over real field

(f) n for complex matrices over complex field and for real matrices over real field, $2n$ for complex matrices over real field.

(g) 1 for complex matrices over complex field, 1 for real matrices over real field and 2 for complex matrices over real field.

26. Prove $U \cap W = U$.

27. Use the fact that $dim(U + W) = dim(U) + dim(W) - dim(U \cap W)$.

28. $U = \{(b_1, b_2, b_3) : b_1 - 5b_2 + 3b_3 = 0\}, W = \{(b_1, b_2, b_3) : b_2 = 0\}$ basis for $U \cap W = \{(3, 0, -1)\}, dim(U + W)$ is trivial

29. Think of a basis for the space from the basis of R^n

30. Find an infinite subset which is linearly independent

31. 3.

32. (a) Basis for $U = \{(1, 2, 3)^T, (2, 1, 1)^T\}$, Basis for $W = \{(1, 0, 1)^T, (3, 0, -1)^T\}$, Basis for $U \cap W = \{(3, 0, -1)^T\}$ Basis for $U + W = \{e_1, e_2, e_3\}$.

    (b) Basis for $U = \{(1, 0, 2, 0)^T, (1, 0, 3, 0)^T\}$, Basis for $W = \{(1, 0, 0, 0)^T, (0, 1, 0, 0)^T, (0, 0, 1, 1)^T\}$ Basis for $U \cap W = \{(1, 0, 0, 0)^T\}$, Basis for $U + W$ is $\{e_1, e_2, e_3, e_4\}$.

    (c) Basis for $U = \{(1, 0, 0, 2)^T, (3, 1, 0, 2)^T, (7, 0, 5, 2)^T\}$ Basis for $W = \{(1, 0, 3, 2)^T, (10, 4, 14, 8)^T$ Basis for $U \cap W = \{(-43/22, -18/11, 15/22, 1)^T\}$ Basis for $U + W = \{(e_1, e_2, e_3, e_4)\}$.

33. (a) It is enough to show that the given set of vectors spans the vectors $v_1, v_2, ..., v_n$.

    (b) Trivial.

34. $b(3 - a) + 6 = 0$.

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